

Introduction

References & abbreviations used

1. **AE-BC** – Aerodynamics for Engineers, (6th edition), John J. Bertin & Russell M. Cummings - [course text](#)
2. **FA-JA** – Fundamentals of Aerodynamics, (6th edition), John D. Anderson Jr.
3. **FA-KC** – Foundations of Aerodynamics, (5th edition), Arnold M. Kuethe & Chuen-Yen Chow
4. **AA-DA** – Applied Aerodynamics: A Digital Text Book - from Desktop Aeronautics Inc. <http://docs.desktop.aero/appliedaero/preface/welcome.html>,
5. **IAFV-JGL** – Introduction to Aerospace Flight Vehicles, J. Gordon Leishman
6. **MFMO** – Multimedia Fluid Mechanics Online, General editor G. M. Homsy
7. **AFM-VD** – Album of fluid motion, Milton Van Dyke
8. **VF-JSME** – Visualized Flow, by The Japan Society of Mechanical Engineers
9. **AE-HCCV** – Aerodynamics for Engineering students, E.L. Houghton, P.W. Carpenter, Steven H. Collicott & Daniel T. Valentine

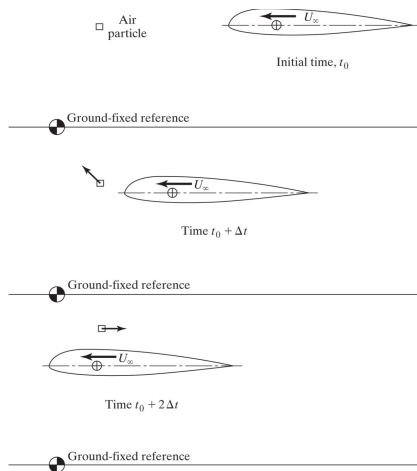
Other abbreviations used

1. **TEq.** - Equation number in text – Aerodynamics for Engineers, (6th edition), John J. Bertin & Russell M. Cummings
2. **TSec** - Section number in text – Aerodynamics for Engineers, (6th edition), John J. Bertin & Russell M. Cummings

Section 1 – Conservation Equations

Reference frame (TSec. 2.1)

Ground-fixed reference frame



► Unsteady (varying with time) flow

► $\frac{\partial()}{\partial t} \neq 0$

Figure 1.1: Fig. 2.1 from AE-BC

Reference frame (TSec. 2.1)

Vehicle-fixed reference frame

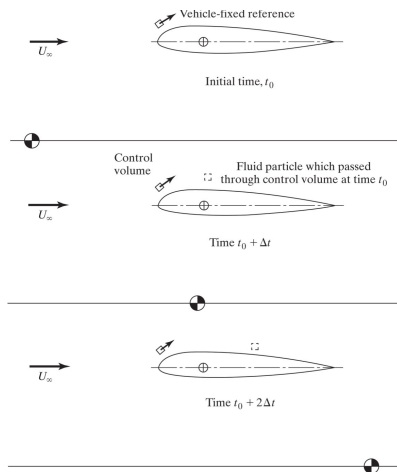


Figure 1.2: Fig. 2.2 from AE-BC

► Steady (does not vary with time) flow

► $\frac{\partial()}{\partial t} = 0$

► Most wind tunnel tests

► Follow in AE 319

Control Volume

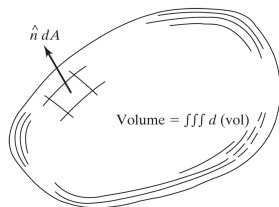


Figure 1.3: Fig. 2.5 from AE-BC

- ▶ Control volume - fixed volume in space through which fluid flows
- ▶ A - surface of the control volume
- ▶ $d(\text{vol})$ - elemental control volume
- ▶ dA - surface of the elemental volume $d(\text{vol})$
- ▶ $\hat{n}$ - unit normal to dA
- ▶ Therefore, Volume = $\iiint d(\text{vol})$

Volumetric flow rate

- ▶ dQ - volume rate of flow through the differential surface dA . Note, $d\vec{A} = \hat{n}dA$
- ▶ units of $dQ \rightarrow \text{m}^3 \text{s}^{-1}$
- ▶ $dQ\Delta t$ - volume of fluid flow through a surface dA in time Δt

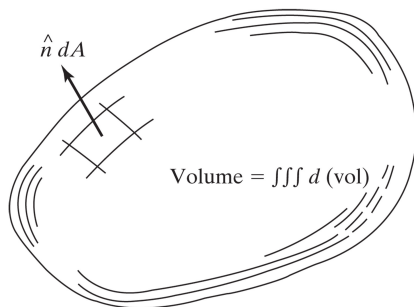
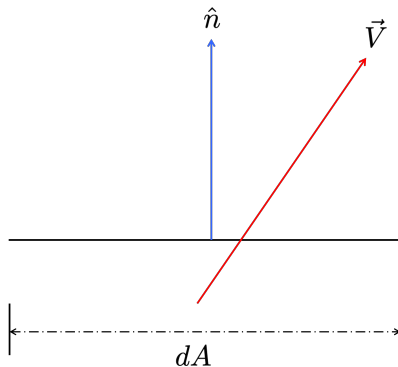


Figure 1.4: Fig. 2.5 from AE-BC

Volumetric flow rate



$$dQ\Delta t = \vec{V} \cdot \hat{n} \Delta t dA \rightarrow \text{volumetric flow in the } \hat{n} \text{ direction} \quad (1.1)$$

Therefore the volumetric flow rate through A is

$$Q = \iint_A dQ = \iint_A \vec{V} \cdot \hat{n} dA = \iint_A (\text{volume flux}) dA \quad (1.2)$$

Mass flow rate

Similarly, the mass flow rate

$$\begin{aligned}\rightarrow \dot{m} &= \iint_S (\text{mass flux}) \, dA \\ \dot{m} &= \iint_A d\dot{m} = \iint_A \rho \times (\text{volume flux}) \, dA \\ \dot{m} &= \iint_A \rho \vec{V} \cdot \hat{n} \, dA\end{aligned}\tag{1.3}$$

if $\rho = \text{constant}$, (incompressible flow),

$$\dot{m} = \rho \iint_A \vec{V} \cdot \hat{n} \, dA = \rho Q\tag{1.4}$$

Conservation of Mass

Statement of conservation of mass: Mass is neither created nor destroyed

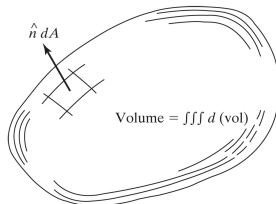


Figure 1.5: Fig. 2.5 from AE-BC

$$\underbrace{\{\text{rate of mass accumulation in volume}\}}_I = \underbrace{\{\text{rate of mass flow into volume}\}}_{II} - \underbrace{\{\text{rate of mass flow out of volume}\}}_{III}$$

Conservation of mass (TSec. 2.2)

$$I \rightarrow \frac{\partial}{\partial t}(\text{mass}) = \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \, d(\text{vol}) \quad (1.5)$$

$$II - III = - \iint_A \rho \vec{V} \cdot \hat{n} \, dA \quad (1.6)$$

Therefore, **global conservation of mass** or the **integral form of the conservation of mass** from eq. (1.5) & (1.6),

$$\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \, d(\text{vol}) + \iint_A \rho \vec{V} \cdot \hat{n} \, dA = 0 \quad (\text{TEq. (2.5)}) \quad (1.7)$$

if the volume is fixed,

$$\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \, d(\text{vol}) = \iiint_{\text{vol}} \frac{\partial \rho}{\partial t} \, d(\text{vol}) \quad (1.8)$$

Conservation of mass

Divergence theorem:

$$\oiint_A \vec{F} \cdot \hat{n} \, dA = \iiint_{\text{vol}} \nabla \cdot \vec{F} \, d(\text{vol}) \quad (1.9)$$

where, $\vec{F}$ is any vector. Using eq. (1.9) in eq. (1.6) we get,

$$\oiint_A (\rho \vec{V}) \cdot \hat{n} \, dA = \iiint_{\text{vol}} \nabla \cdot (\rho \vec{V}) \, d(\text{vol}) \quad (1.10)$$

Substituting eq. (1.10) in eq. (1.7) we get,

$$\iiint_{\text{vol}} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) \right] d(\text{vol}) = 0 \quad (1.11)$$

The volume is fixed, but it has been chosen **arbitrarily** hence it has to be true **everywhere** .

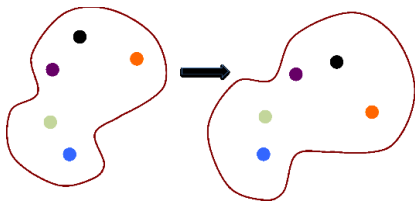
Conservation of mass

We can also shrink the volume to a point (every point in a flow field).
For eq. (1.11) to be true then,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \quad (\text{TEq. (2.2)}) \quad (1.12)$$

everywhere, which is known as the **local conservation of mass** or the **differential form of the conservation of mass**.

Divergence of velocity



Consider a control volume (vol) with fluid particles (colored points), at two instances in time, shown above. This control volume can remain unchanged, expand, or contract. We can mathematically represent this by the divergence of the velocity field of the fluid particles within the control volume. The divergence of the velocity field in 2-D cartesian coordinates is,

$$\nabla \cdot \vec{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \quad (1.13)$$

The divergence of the velocity field $\nabla \cdot \vec{V}$ is the **rate of change of volume of a moving fluid element per unit volume**.

Divergence of velocity

- ▶ if $\nabla \cdot \vec{V} = 0$, then the velocity is *divergence free* or solenoidal. This implies that the volume of the infinitesimal fluid element is unchanged and the fluid flow in the control volume under consideration is **incompressible**.
- ▶ if $\nabla \cdot \vec{V} < 0$, then the fluid flow in the infinitesimal control volume under consideration is **contracting**.
- ▶ if $\nabla \cdot \vec{V} > 0$, then the fluid flow in the infinitesimal control volume under consideration is **expanding**.

Conservation of Momentum (TSec. 2.3)

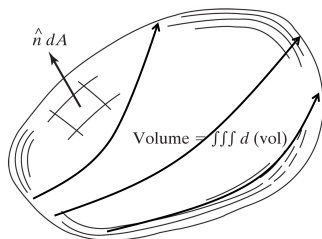


Figure 1.6: modified Fig. 2.5 from AE-BC

State of conservation of momentum: Newton's second law.
Mathematically expressed as

$$\vec{F} = m\vec{a} = \frac{d}{dt}(m\vec{V}) \quad (1.14)$$

where $\vec{F}$ is the force on the fluid, $\vec{a}$ the acceleration and m the mass.

Conservation of Momentum

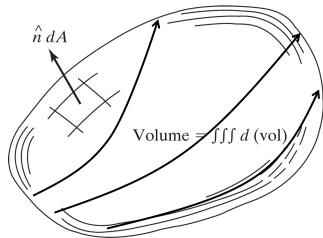


Figure 1.7: modified Fig. 2.5 from AE-BC

$$\begin{aligned} & \{\text{rate of momentum accumulation in (vol)}\} \\ &= \{\text{rate of momentum in}\} - \{\text{rate of momentum out}\} + \\ & \quad \{\text{sum of forces acting on the system}\} \\ &= -\{\text{net rate of momentum flow out of (vol)}\} \\ & \quad + \{\text{sum of forces acting on the system}\} \end{aligned} \quad (1.15)$$

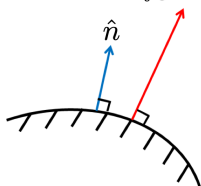
Applied forces

- ▶ Body forces - acts throughout (vol), e.g., gravity, electromagnetic forces
- ▶ Surface forces - acts only on A, e.g., pressure, viscous forces

$$\{\text{body force on (vol)}\} = \vec{F}_{\text{body}} = \iiint_{(\text{vol})} \rho \vec{F}_b d(\text{vol}) \quad (1.16)$$

where, ρ is mass/unit volume and $\vec{F}_b$ is force/unit area.

$$\{\text{surface force on (vol)}\} = \vec{F}_{\text{surface}} = \oint_A f_s \hat{n} \cdot dA \quad (1.17)$$



Conservation of momentum

Using eq. (1.16) and (1.17) in eq. (1.15) we get,

$$\underbrace{\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \vec{V} \, d(\text{vol})}_{\text{accumulation}} = - \underbrace{\oint_A \rho \vec{V}}_{\text{I}} \underbrace{(\vec{V} \cdot \hat{n})}_{\text{II}} \, dA - \oint_A p \hat{n} \cdot dA + \vec{F}_{visc} + \vec{F}_{ext} + \iiint_{\text{vol}} \rho \vec{f}_b \, d(\text{vol}) \quad (1.18)$$

where, I - quantity being transported, momentum ($\rho \vec{V}$) and II - velocity at which quantity is being transported. Here the momentum is being transported at a velocity $\vec{V} \cdot \hat{n}$. Also, p is the pressure, $\vec{F}_{visc}$ is the viscous forces, $\vec{F}_{ext}$ is any other external force that may be acting, and $\vec{f}_b$ (per unit volume) is any body force that may be acting. This is the **global** or **integral** form of the momentum equations. It's a vector equation with three equations when considering 3-D flow.

Conservation of momentum

Your text has the global or integral form of the momentum in the following form,

$$\vec{F}_{\text{body}} + \vec{F}_{\text{surface}} = \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \vec{V} \, d(\text{vol}) + \iint_A \vec{V} (\rho \vec{V} \cdot \hat{n} \, dA) \quad (\text{TEq. (2.13)}) \quad (1.19)$$

Here, the pressure forces has been lumped into $\vec{F}_{\text{surface}}$

Conservation of momentum

Following the procedure we used for the conservation of mass we utilize the divergence theorem. We then can shrink (vol) to a point and we know that the choice of the control volume (vol) is arbitrary so we get at every point in the flow the following mathematical statement of the conservation of momentum in the differential form (local form).

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \vec{f} + \frac{1}{\rho} (\vec{F}_{visc} + \vec{F}_{ext}) \quad (1.20)$$

This is the **local** or **differential** form of the momentum equations. It's a vector equation with three equations when considering 3-D flow.

This equation in its expanded form is presented in your text in
(TEq. (2.12))

Similarity - AE 201 Revision+ (TSec. 2.5)

Similarity:

Scale model and full scale will have identical non-dimensional numbers. IF

- ▶ They are geometrically similar (scale model)
- ▶ They are dynamically similar. For most flows in AE 319, dynamic similarity is achieved by matching
 - ▶ Reynolds $\# = \frac{\rho V L}{\mu}$
 - ▶ Mach $\# = \frac{V}{a}$

This can be rigorously shown using the differential form of the momentum equation. This is beyond the scope of AE 319 but is derived in your textbook in Sec 2.5. To state the derivation in words,

- ▶ Non-dimensionalize the momentum equation
- ▶ It will be concluded that if the Reynolds number and Mach number of the scale model and the full-scale object (say airfoil) are matched then the momentum equation is exactly the same.

Reynolds number regimes

TABLE 2.2 Reynolds Number Regimes and Characteristics for an Airfoil

<i>Reynolds number regime</i>	<i>Reynolds number range</i>	<i>Characteristics</i>
Creep	$0 < Re_L < \approx 10^2$	Viscous effects dominate, drag inversely proportional to Re_L
Low Re_L laminar	$\approx 10^2 < Re_L < \approx 10^4$	Vortex shedding, separation bubbles, reduced lift, drag proportional to Re_L
Laminar	$\approx 10^4 < Re_L < \approx 10^5$	Separation bubbles, decreased skin friction, earlier separation, earlier stall
Transitional	$\approx 10^5 < Re_L < \approx 10^6$	Both laminar and turbulent boundary layers with characteristics of both
Turbulent	$Re_L > \approx 10^6$	Inertial effects dominate, increased skin friction, delayed separation, delayed stall

Figure 1.8: Table. 2.2 from AE-BC

Mach number regimes

TABLE 2.1 Mach Number Regimes and Characteristics for an Airfoil

<i>Mach regime</i>	<i>Mach number range</i>	<i>Characteristics</i>
Incompressible Subsonic	$0 < M < \approx 0.3$	Classical low speed aerodynamics
Compressible Subsonic	$\approx 0.3 < M < \approx 0.8$	Compressibility effects, no shocks
Transonic	$\approx 0.8 < M < \approx 1.6$	Normal shocks, shock/boundary layer interactions, etc.
Supersonic	$\approx 1.6 < M < \approx 5.0$	Oblique shocks, expansion waves, linear characteristics
Hypersonic	$M > \approx 5.0$	Bow shocks, thin shock layers, high temperature effects, shock/shock interactions, low density flows, etc.

Figure 1.9: Table. 2.1 from AE-BC

Conservation of energy (TSec. 2.7)

Statement of conservation of energy - first law of thermodynamics i.e., energy in an isolated system cannot be created or destroyed (it is constant) it can only be transformed from one form to another. For a **closed system**,

$$\{\text{change in energy}\} = \{\text{work done on the system}\} + \{\text{heat transfer to the system}\} \quad (1.21)$$

$$\rightarrow dE = dW + dQ \quad (1.22)$$

where, E is energy, W is work done and Q is heat transfer. Your textbook (TEq. (2.2)) uses lower case e , w and q . This equation (TEq. (2.2)) in your textbook is in terms of per unit mass, hence the use of lowercase letters.

Conservation of energy

Considering an **open system**,

$$\begin{aligned} & \{\text{rate of change in total energy}\} = \\ - & \{\text{net rate of change of total energy flow } \underline{\text{out}} \text{ of (vol) by convection}\} \\ & + \{\text{net rate of work done } \underline{\text{on}} \text{ (vol) by the surroundings}\} \\ & + \{\text{net rate of heat added } \underline{\text{to}} \text{ (vol) other than by convection}\} \end{aligned} \quad (1.23)$$

Total energy

Total energy = internal energy + kinetic energy + potential energy

$$E_t = \underbrace{\rho u_e}_{\text{internal energy}} + \underbrace{\frac{1}{2}\rho[u^2 + v^2 + w^2]}_{\text{kinetic energy}} + \underbrace{\rho g z}_{\text{potential energy}} \quad (1.24)$$

$$E_t = \rho e + \frac{1}{2}\rho \vec{V} \cdot \vec{V} + \rho g z \quad (1.25)$$

where, u_e is the internal energy per unit mass, u, v & w are the x, y & z components of $\vec{V}$, g is the acceleration due to gravity and z the ordinate in the direction in which the potential energy is changing (height).

Total energy

Therefore,

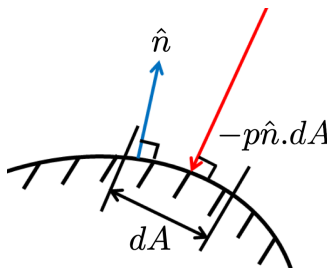
$$\{\text{rate of change in } E_t\} = \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) d(\text{vol}) \quad (1.26)$$

$$\begin{aligned} \{\text{rate of change of } E_t \text{ flow } \underline{\text{out}} \text{ of (vol) by convection}\} = \\ \oiint_A \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) \vec{V} \cdot \hat{n} dA \quad (1.27) \end{aligned}$$

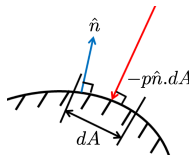
Work done by forces

$$\text{work} = \text{force} \times \text{distance} \quad (1.28)$$

$$\text{rate of work} = \text{force} \cdot \frac{\text{distance}}{\text{time}} = \text{force} \cdot \text{velocity} = \vec{F} \cdot \vec{V} \quad (1.29)$$



Work done by forces



$$\{\text{net rate of } W \text{ done by } p\} = - \iint_A p \vec{V} \cdot \hat{n} dA \quad (1.30)$$

$$\{\text{net rate of } W \text{ done by body forces e.g., gravity}\} = \iiint_{\text{vol}} \rho \vec{f}_b \cdot \vec{V} d(\text{vol}) \quad (1.31)$$

$$\{\text{net rate of } W \text{ done by viscous forces}\} = \dot{W}_{visc} \quad (1.32)$$

$$\{\text{Shaft work}\} = \dot{W}_s \quad (1.33)$$

Here shaft work is work done through a shaft e.g., turbine, pump

Modes of heat transfer

- ▶ **convective** - due to the flow of fluid into and out of (vol) already included
- ▶ **conductive** - due to random molecular motion carrying heat.
What feels colder 45°F water or air?
- ▶ **radiative** - due to absorption and emission of electromagnetic radiation. Neglect in AE319. Important in combustion problems, re-entry vehicles, satellites etc.

$$\{\text{rate of heat transfer by conduction}\} = \dot{Q} \quad (1.34)$$

Conservation of energy

As before, **global or the integral form** of the conservation of energy becomes,

$$\begin{aligned} & \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \left(u_e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + gz \right) d(\text{vol}) = \\ & - \iint_A \rho \left(u_e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + gz \right) \vec{V} \cdot \hat{n} dA \\ & + \dot{Q} - \iint_A p \vec{V} \cdot \hat{n} dA + \dot{W}_{visc} + \dot{W}_s + \iiint_{\text{vol}} \rho \vec{f}_b \cdot \vec{V} d(\text{vol}) \end{aligned} \quad (1.35)$$

Note, the equivalent equation in your textbook (TEq. (2.37)) is in a slightly different form. The differences are highlighted in the following slide.

Conservation of energy

- ▶ Your textbook combines the net rate of work done by pressure with the convection terms $((\quad)\vec{V} \cdot \hat{n})$. Here, it is kept separate to be consistent with the momentum equation. This is also intuitive from a physics standpoint as any work done by the pressure forces will result in a change in the convective terms.
- ▶ In eq. (1.35) an explicit work done by body forces is retained, electromagnetic for example.

Conservation of energy

Using the same procedure as before (divergence theorem etc.) the **local or differential form** of the conservation of energy becomes,

$$\frac{\partial}{\partial t} \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) + \nabla \cdot \left[\rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) \vec{V} \right] = \dot{q} - \nabla \cdot (p \vec{V}) + \dot{w}_{visc} + \rho \vec{f}_b \cdot \vec{V} \quad (1.36)$$